The Yangtze River

by Gary Arndt

Subscribe
Apple | Spotify | Amazon | Player FM | Tunein
Castbox | Podurama | Podcast Republic | RSS | Patreon

Podcast Transcript

Located within the nation of China is the world's third-longest river, the Yangtze.

The Yangtze River has been one of the defining geographical features in China since the dawn of history. It had a significant role in cultivating Chinese civilization, and its flooding has been responsible for some of the greatest disasters in the world.

Controlling the power of the river has resulted in the construction of the world's largest dam.

Learn more about the Yangtze River, one of the great rivers of the world, on this episode of Everything Everywhere Daily.

Each of the major rivers in the world has a different story. There are some rivers, like the Nile and the Tigris and Euphrates, which have very deep histories and were fundamental in the rise of the civilizations on their banks.

There are other rivers, like the Rhine and the Mississippi, which don't have quite the same historical credentials but are extremely important economically.

The Yangtze River checks both boxes. It has deep historical significance, and it remains one of the most important economic rivers in the world today.

Before I get into the history and importance of the river, I should explain some of the geographical facts of the river.

The Yangtze is the world's third-largest river by length, behind only the Nile and the Amazon, and it is the fifth or sixth-largest river in the world in terms of total discharge volume, depending on how you define it and the season.

It is also the longest river in the world, which is entirely in a single country, and the longest river in the Eurasian landmass.

The river begins in the Tibetan Plateau and winds 6,300 kilometers or approximately 3,915 miles to the Pacific Ocean, near Shanghai.

The Yangtze River basin is the heart of China, covering a full one-fifth of the area of the country and one-third of the population. The Yangtze Delta alone, near Shanghai, is responsible for a fifth of the Chinese economy.

The name Yangtze is actually not the name of the river in Mandarin. The word "Yangtze' is actually an anglicization of the French term for the river. In Chinese, it is known as Cháng Jiāng, which simply means "long river."

The river flows roughly from west to east, and it has traditionally served as the dividing line between northern and southern China, similar to how the Mississippi divides the United States between east and west.

The river can be roughly divided into three sections: the upper, middle, and lower Yangtze.

The upper Yangtze begins in the Tanggula Mountains on the Qinghai-Tibetean Plateau and flows eastward through the Qinghai, Tibet, and Yunnan provinces. This stretch of the river is known for its deep gorges and valleys.

The middle Yangtze flows through the heart of China. This is the most developed and economically important section of the river. It flows through the provinces of Sichuan, Chongqing, Hubei, Hunan, and Jiangxi, and it is home to many of China's largest cities, including Chongqing, Wuhan, and the former imperial capital of Nanjing. The Yangtze Basin is often referred to as the "breadbasket of China" due to its fertile farmlands.

The middle Yangtze is also the location of the Three Gorges. It is a series of three gorges which, for much of history, served as a limit for navigation on the river.

The lower Yangtze constitutes the Yangtze River delta, which is roughly defined from Nanjing to the sea.

Evidence of humans along the Yangtze goes back a long time. The first archeological evidence of human activity dates back 27,000 years in the Three Gorges area.

Despite the important role that the Yangtze has played throughout Chinese history, what we think of as Chinese culture and civilization today did not actually originate along the Yangtze. It originated along the Yellow River to the north.

About 7,000 years ago, the lower Yangtze became a major population center with people known as the Yue. The Yue had very different cultural practices than the people in northern China, including dying their teeth black and heavy tattooing.

After the defeat of Yue by the Qing Dynasty, they began to adopt more of the cultural practices of the north.

After the consolidation of China, the importance of the Yangtze only increased. Irrigation systems were put in place during the Han Dynasty, and by the Song Dynasty, the Yangtze had become the wealthiest region in China.

In addition to being agriculturally productive, the Yangtze was also a major transportation corridor, allowing goods to be transported from the sea all the way into the interior of China. It was part of the many routes which compromised the Silk Road.

The importance of the Yangtze was amplified with the construction of the Chinese Grand Canal. While Nanjing was the capital of China for several periods in Chinese history, so was Beijing. The Grand Canal connected Beijing with the Yangtze River.

The Grand Canal is probably worth an episode of its own at some point in the future, but it is the longest artificial waterway in the world. Construction began on it in the 4th century BC, and construction continued for centuries. The total length of the canal is approximately 1,100 miles or 1,800 kilometers.

As productive and fertile as the Yangtze River valley was, there was a problem. Flooding.

The extensive irrigation system, which was built over the centuries, did mitigate much of the flooding on the Yangtze, but it didn't eliminate it. Occasionally, flooding would be truly devastating.

The river's worst flood, and what was probably the worst flood in world history, took place along the Yangtze in 1931.

China experienced a severe drought from 1928 to 1930. The winter of 1930-31 saw unusually heavy snowfall in the Tibetan plateau.

As the snow melt started in the spring, it coincided with extremely strong spring rains.

By June, low-lying regions along the Yangtze were already underwater.

The spring rains were just the start of an extremely long rainy season, which also saw an unusual nine typhoons make landfall that year.

The result was a flooding of not just the Yangtze but many of the tributaries of the river in the entire drainage basin.

On top of the extreme rainfall, the dykes and levees on the Yangtze were in poor shape and were unable to hold back the flood waters.

The result was the flooding of 180,000 square kilometers or 69,000 square miles of land. This is the equivalent of most of the island of Great Britain or the states of New York, New Jersey, and Connecticut combined.

Farm fields were flooded, which compounded the problem of the drought from the previous years.

The number of fatalities from the flooding has been estimated to be as high as 4,000,000 people. The 1931 flood ranks not just as the worst flood in history in terms of total death toll but also as the worst natural disaster of any type in history.

The Yangtze experienced severe flooding in 1954 and again in 1998. The flooding wasn't frequent, but when it happened, it could be devastating.

The idea of controlling flooding on the Yangtze goes back to the early 20th century. The founder of the Republic of China, Doctor Sun Yat Sen, on whom I've done a previous episode, proposed the creation of a dam on the Yangtze.

In addition to controlling flooding, such a dam would also provide an enormous amount of electricity for China.

An initial survey was conducted near the Three Gorges section of the river in 1932, immediately after the 1931 flooding. In 1944, during the war, United States engineers created a plan, and they even brought Chinese engineers back to the United States for training, but the plan never came to fruition due to the Nationalists losing the Chinese Civil War.

The Communists under Mao Zedong supported the idea of a Yangtze dam, but they were unable to make any progress on it. The Great Leap Forward and the Cultural Revolution left China in a position where they were unable to undertake such a large project.

While a dam on the Yangtze would control flooding and produce an enormous amount of electrical power, there were serious environmental and cultural concerns about what a dam would do.

The lake created by a dam would destroy many communities, cultural sites, and well as plant and animal species that inhabited the river.

In the 1980s, with Mao behind them, the Chinese government ignored most of the concerns and decided to proceed with the dam. The dam project was approved at the 1992 session of the National People's Congress, with 67% support.

I mention the vote only because this was an extremely low percentage of support for a body that normally voted on things unanimously.

Construction began in 1994, and the dam was expected to be completed in 2006, but the project wasn't completed until 2012, almost eighteen years after construction began.

Dubbed the Three Gorges Dam, it is the largest dam in the world. It is 2.3 kilometers or 1.5 miles long. The damn used 27.2 million cubic meters of cement and enough steel to build 63 Eiffel Towers.

The amount of electricity generated by the dam is astonishing. When all of its turbines are functioning, it can produce 22,500 MW of power. It can produce more than twice the amount of electricity as the world's next largest damn.

It was estimated that when the dam produced a thousand terawatt hours of electricity, it would pay for the construction of the project. It was estimated that it would take ten years to recoup the investment once the dam was completed.

The dam produced a thousand terawatt hours of power within its first year.

In addition to the dam, there were also multiple locks constructed to facilitate transportation on the river. Before the dam's construction, the Three Gorges was very difficult and dangerous to navigate.

The new locks significantly reduced the cost of transportation on the river and increased the amount of freight transported on the river by almost an order of magnitude.

Prior to the construction of the dam, 18.0 million tonnes of cargo passed through the Three Gorges area per year. In 2022, there were over 159.65 million tons.

The Yangtze has a lot of problems due to the rapid growth of China over the last several decades. The Yangtze River sends more plastic pollution into the ocean than any other river in the world by a wide margin.

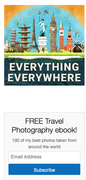
90% of all the plastic in the ocean comes from just ten rivers, and the Yangtze has more plastic waste than the other nine rivers in the top 10 combined.

The good news is that steps are being taken to reduce the amount of plastic pollution in the river, which would be the single biggest thing that could be done on Earth to reduce overall plastic pollution.

The Yangtze has been one of the most important rivers in the world for thousands of years and it remains so today, being a source of transportation, electricity, and irrigation for much of China.

Whatever the future may hold, it is quite likely that the Yangtze will remain central to China for centuries to come.

Everything Everywhere Daily
All About Oman
What is Calculus?



FREE Travel Photography ebook!
100 of my best photos taken from around the world
Email Address
Subscribe
We won't send you spam. Unsubscribe at any time.

© 2006-2024 Everything Everywhere Travel Media